

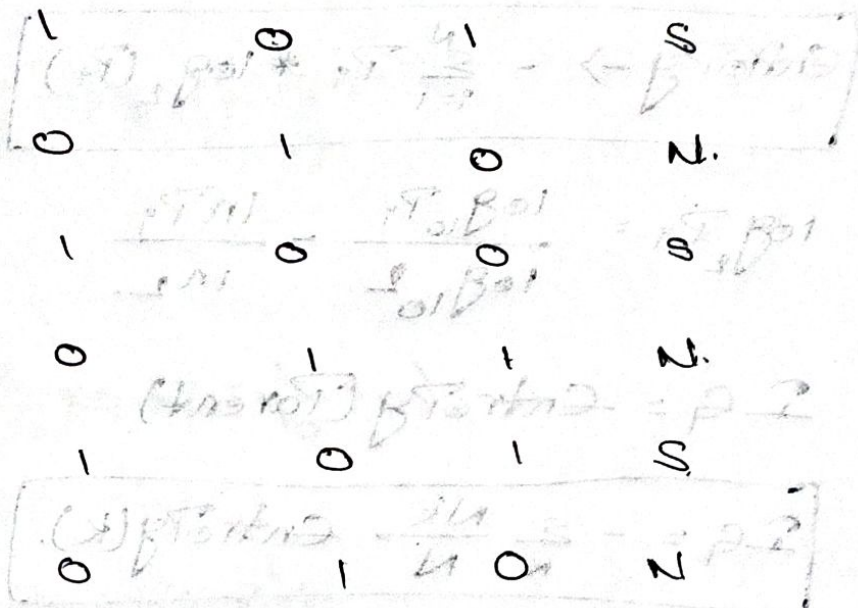
Tutorial - 8

24/03/2026.

Decision tree

Email ID Suspicious Words Money Keywords Sender known Has Attachment Class.

1	8	1	0	1	S
2	1	0	1	0	N
3	6				
4	2				
5	7				
6	1				
7	9				
8	2				
9	5				
10	1				



Questions:

- Which attribute should be selected as the root node? Justify your answer with calculations.
- Build the complete Decision tree using the ID3 algorithm (Use Information Gain). Draw the tree structure with all nodes, branches, and leaf nodes clearly labeled.
- What is the depth of your decision tree? How many leaf nodes does it have?

ID₃ → Entropy & Information Gain.

(IG) → Information Gain of each feature.

IG → max → best feature.

Goal: Reduce Entropy.

1. Increase Information Gain.

$$\text{Entropy} \rightarrow - \sum_{i=1}^N P_i * \log_2(P_i)$$

$$\log_2 P_i = \frac{\log_{10} P_i}{\log_{10} 2} = \frac{\ln P_i}{\ln 2}$$

$$IG = \text{Entropy}(\text{Parent})$$

$$IG = - \sum_K \frac{N_K}{N} \text{Entropy}(K)$$

① Entropy (Parent & y) =

$$P_{\text{spam}} = \frac{5}{10} = 0.5$$

$$P_{\text{non-spam}} = \frac{5}{10} = 0.5$$

$$\text{Entropy} = - [0.5 \log_2(0.5) + 0.5 * \log_2(0.5)]$$

$$= 1$$

② Entropy (Money keywords)

1 → spam.

→ non-spam(x)

} Perfectly
classified
so Entropy = 0.

0 → non-spam.

→ spam(x)

$$IG(m.k) = 1 - 0 = 1$$

③ Entropy (sender known)

1 → Spam(1)

→ non Spam(5)

0 → Spam(4)

→ non Spam(0).

$$\text{Entropy} = - \left[\frac{1}{6} \log_2 \left(\frac{1}{6} \right) + \frac{5}{6} \log_2 \left(\frac{5}{6} \right) \right]$$

$$\text{for '0' } \rightarrow 0 + \frac{4}{4} \log_2 \left(\frac{4}{4} \right)$$

$$= 0.650$$

$$\approx 0.65$$

$$IG(S.K) = 1 - \left(\frac{6}{10} \times 0.65 \times \frac{4}{10} \times 0 \right)$$

$$= 0.61$$

④ Entropy (has attachments):-

0 — Spam = 2 } Entropy > 0.
 ↑ — non-Spam = 3 }

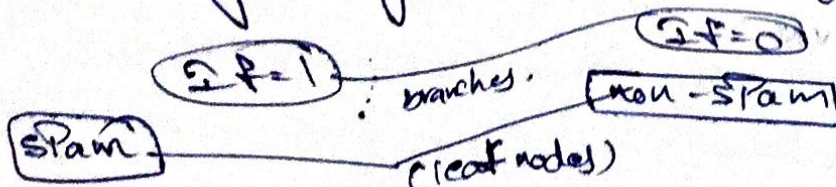
1 — Spam = 3 } Entropy > 0.
 — non-Spam = 2 }

⑤ Entropy (suspicious words):-

Entropy > 0 (Random value in dataset).

$$IG < 1$$

Money Keywords: Root Node.



Depth = 1 (only one decision node is there)

leaf nodes = 2